Project: Retrieval Augmented Generation Chatbot

This project is a RAG chatbot. The dataset we use comes from the *arXiv* website. We have selected a few research articles about the impact of generative AI on humans and labor. We will download and store them in a volume in *Chapter 2*. After we download the PDF documents, we will extract and prepare the text through a process of chunking and tokenization, creating embeddings of the documents to be referenced in the chatbot. We will use Databricks Vector Search to store the embeddings. Then, we will use the new Foundation Model API to generate answers when text is retrieved. The final bot will be deployed as an application using **Databricks Model Serving**. This example allows you to build a chatbot from start to finish! Take a look through the project pipeline in *Figure 1.10* to understand how we'll progress through this project.

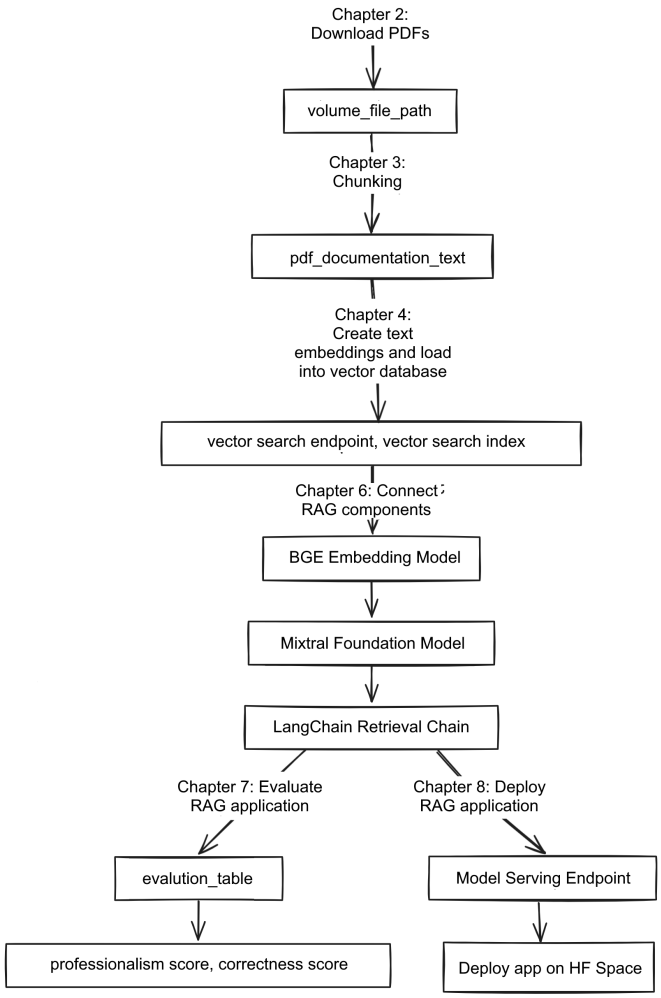


Figure 1.10 – The project pipeline for the chatbot project